SUMS OF POWERS VIA BACKWARD FINITE DIFFERENCES AND NEWTON'S FORMULA

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Abstract. In this manuscript, we derive closed formulas for multifold sums of powers of integers by combining the backward Newton interpolation formula with hockey-stick identities for binomial coefficients. We further obtain representations of multifold sums of powers in terms of Stirling numbers of the second kind and Eulerian numbers. Finally, we provide Wolfram Mathematica programs for the efficient verification of the derived identities.

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1. Auxiliary Lemmas

In this section, we introduce several lemmas, definitions, and conventions that serve as the foundation for the main results of this manuscript. For multifold sums of powers, we use the notation provided by Donald Knuth in [1], which is concise and useful.

Definition 1.1 (Multifold Sums of Powers)**.** *For non-negative integers r, n, and m, the multifold sums are defined recursively by*

$$\Sigma^0 n^m = n^m,$$

$$\Sigma^1 n^m = \Sigma^0 1^m + \Sigma^0 2^m + \cdots + \Sigma^0 n^m,$$

$$\Sigma^{r+1} n^m = \Sigma^r 1^m + \Sigma^r 2^m + \cdots + \Sigma^r n^m.$$

We use the following notation for rising factorials.

Definition 1.2 (Rising Factorials)**.** *For integers n and k, define the rising factorial by*

$$n^{(k)} = \begin{cases} 0, & \text{if } k < 0, \\ 1, & \text{if } k = 0, \cdot \\ \prod_{j=0}^{k-1}(n+j), & \text{if } k > 0. \end{cases}$$

For further discussion of generalized factorials, see J. F. Steffensen [2].

Lemma 1.3 (Backward Newton's formula)**.** *Let $f\colon \mathbb{Z} \to \mathbb{C}$ be a function, and let $t \in \mathbb{Z}$. Then, for all $n \in \mathbb{Z}$,*

$$f(n) = \sum_{k=0}^{\infty} \frac{(n-t)^{(k)}}{k!} \nabla^k f(t) = \sum_{k=0}^{\infty} \binom{n-t+k-1}{k} \nabla^k f(t),$$

Here, $\nabla^k f(t)$ denotes the k-th backward difference of f evaluated at t

$$\nabla^k t^m = \sum_{j=0}^{k} (-1)^j \binom{k}{j} (t-j)^m.$$

Proof. Sir Isaac Newton originally presented this formula in [3]. J. F. Steffensen expressed it in modern mathematical notation in [4, Section 2, Equation (19)]. The identity $\frac{(n-t)^{(k)}}{k!} = \binom{n-t+k-1}{k}$ gives the binomial form. □

Lemma 1.4 (Backward Newton's formula for powers)**.** *For non-negative integers n and m, and an arbitrary integer t, the following identity holds*

$$n^m = \sum_{k=0}^{m} \binom{n-t+k-1}{k} \nabla^k t^m.$$

Proof. This is the special case of Lemma (1.3) for powers. □

Lemma 1.5 (Generalized Hockey-Stick Identity)**.** *For arbitrary integers a, b, and j, the following identity holds*

$$\sum_{k=a}^{b} \binom{k}{j} = \binom{b+1}{j+1} - \binom{a}{j+1}.$$

Proof. We have $\sum_{k=a}^{b} \binom{k}{j} = \binom{a}{j} + \binom{a+1}{j} + \cdots + \binom{b}{j} = \left(\sum_{k=0}^{b} \binom{k}{j}\right) - \left(\sum_{k=0}^{a-1} \binom{k}{j}\right)$. The hockey-stick identity $\sum_{k=0}^{n} \binom{k}{j} = \binom{n+1}{j+1}$ therefore gives

$$\sum_{k=a}^{b} \binom{k}{j} = \left(\sum_{k=0}^{b} \binom{k}{j}\right) - \left(\sum_{k=0}^{a-1} \binom{k}{j}\right) = \binom{b+1}{j+1} - \binom{a}{j+1}.$$

This completes the proof. □

Lemma 1.6 (Backward Hockey-Stick Identity)**.** *For an integer $n \geq 1$ and arbitrary integers t, k, and r, the following identity holds*

$$\sum_{j=1}^{n} \binom{j-t+k-1+r}{k+r} = \binom{n-t+k+r}{k+r+1} - \binom{k-t+r}{k+r+1}.$$

Proof. Setting $a = k - t + r$ and $b = n - t + k - 1 + r$ in Lemma (1.5) gives the result. □

Lemma 1.7 (Multifold Sums of Zero Powers)**.** *For non-negative integers r and n, the following identity holds*

$$\Sigma^r n^0 = \binom{r+n-1}{r}.$$

Proof.

(1) When $r = 0$, Definition 1.1 gives $\Sigma^0\, n^0 = 1$.

(2) When $r = 1$, we have $\Sigma^1\, n^0 = \sum_{k=1}^{n} 1 = \binom{n}{1}$.

(3) When $r = 2$, we have $\Sigma^2\, n^0 = \sum_{k=1}^{n} \binom{k}{1} = \binom{n+1}{2}$.

(4) When $r = 3$, we have $\Sigma^3\, n^0 = \sum_{k=1}^{n} \binom{k+1}{2} = \binom{n+2}{3}$.

(5) For arbitrary non-negative integer r, iterating the definition and applying the hockey-stick identity gives

$$\begin{aligned}\Sigma^r\, n^0 &= \sum_{k=1}^{n} \Sigma^{r-1}\, k^0 \\ &= \sum_{k=1}^{n} \binom{k+r-2}{r-1} \\ &= \binom{r+n-1}{r}.\end{aligned}$$

This completes the proof. □

Lemma 1.8 (Upper Binomial Negation)**.** *For integers n and k, the following identity holds*

$$\binom{-n}{k} = (-1)^k \binom{k+n-1}{k}.$$

Proof. See [5, p. 164, Equation (5.14)]. □

Convention 1.9. *For every x, we use the convention*

$$x^0 = 1.$$

Donald Knuth gives an extensive discussion of the convention $x^0 = 1$ for all x, including zero, in Concrete Mathematics [5, p. 162] and in [6].

2. Introduction

In this manuscript, we derive formulas for sums of powers by combining backward Newton's interpolation formula for powers in Lemma (1.4) with the hockey-stick identity in Lemma (1.6). We implement the algorithm for finding closed forms of sums of powers of integers, proposed in [7, Alg. (11.2)]. More precisely, the main idea is to identify the binomial

basis of Newton's formula for several difference operators. The corresponding variants are

$$\begin{cases} f(x) &= \sum_{k=0}^{\infty} \frac{(x-a)^{[k]}}{k!} \delta^k f(a) = \sum_{k=0}^{\infty} \frac{x-a}{k} \binom{x-a+\frac{k}{2}-1}{k-1} \delta^k f(a), \\ f(x) &= \sum_{k=0}^{\infty} \frac{(x-a)_k}{k!} \Delta^k f(a) = \sum_{k=0}^{\infty} \binom{x-a}{k} \Delta^k f(a), \\ f(x) &= \sum_{k=0}^{\infty} \frac{(x-a)^{(k)}}{k!} \nabla^k f(a) = \sum_{k=0}^{\infty} \binom{x-a+k-1}{k} \nabla^k f(a), \\ f(x) &= \sum_{k=0}^{\infty} \frac{(x-a)^k}{k!} \frac{\mathrm{d}^k f(a)}{\mathrm{d}x^k}. \end{cases}$$

Each variation of Newton's formula above has its binomial basis:

$$\begin{cases} \frac{(x)_n}{n!} &= \frac{1}{n!} x(x-1)(x-2)\cdots(x-n+1) = \binom{x}{n}, \\ \frac{x^{(n)}}{n!} &= \frac{1}{n!} x(x+1)(x+2)\cdots(x+n-1) = \binom{x+n-1}{n}, \\ \frac{x^{[n]}}{n!} &= \frac{1}{n!} \left(n+\frac{k}{2}-1\right)\left(n+\frac{k}{2}-2\right)\cdots\left(n-\frac{k}{2}+1\right) = \frac{x}{n}\binom{x+\frac{n}{2}-1}{n-1}. \end{cases}$$

These bases allow us to combine Newton's formula with hockey-stick-type identities to find closed forms for sums of powers of integers. Accordingly, the main results use backward Newton's formula and the hockey-stick identity in Lemma (1.6).

3. Main Results

We begin with Lemma (1.4), which gives backward Newton's interpolation formula for powers. For example, the formula gives the following expansions for cubes:

$$n^3 = 0\binom{n-1}{0} + 1\binom{n}{1} - 6\binom{n+1}{2} + 6\binom{n+2}{3},$$

$$n^3 = 1\binom{n-2}{0} + 1\binom{n-1}{1} + 0\binom{n}{2} + 6\binom{n+1}{3},$$

$$n^3 = 8\binom{n-3}{0} + 7\binom{n-2}{1} + 6\binom{n-1}{2} + 6\binom{n}{3},$$

$$n^3 = 27\binom{n-4}{0} + 19\binom{n-3}{1} + 12\binom{n-2}{2} + 6\binom{n-1}{3}.$$

The transition to a formula for sums of powers is straightforward:

$$\Sigma^1 n^m = \sum_{k=0}^{m} \left[\sum_{j=1}^{n} \binom{j-t+k-1}{k} \right] \nabla^k t^m.$$

Setting $r = 0$ in Lemma (1.6) therefore gives the following closed form for ordinary sums of powers.

Proposition 3.1 (Ordinary Sums of Powers)**.** *For non-negative integers n and m, and an arbitrary integer t, the following identity holds*

$$\Sigma^1 n^m = \sum_{k=0}^{m} \left[\binom{n-t+k}{k+1} - \binom{k-t}{k+1} \right] \nabla^k t^m.$$

The basis $\binom{j-t+k-1+r}{k+r}$ is well suited to repeated applications of Lemma (1.6). Indeed, setting the parameter r gives a closed form for sums of powers. For example, the case of cubes is as follows:

Example 3.2. *For integers t with $0 \le t \le 3$, we have*

$$\Sigma^1 n^3 = 0\left[\tbinom{n}{1} - \tbinom{0}{1}\right] + 1\left[\tbinom{n+1}{2} - \tbinom{1}{2}\right] - 6\left[\tbinom{n+2}{3} - \tbinom{2}{3}\right] + 6\left[\tbinom{n+3}{4} - \tbinom{3}{4}\right],$$

$$\Sigma^1 n^3 = 1\left[\tbinom{n-1}{1} - \tbinom{-1}{1}\right] + 1\left[\tbinom{n}{2} - \tbinom{0}{2}\right] + 0\left[\tbinom{n+1}{3} - \tbinom{1}{3}\right] + 6\left[\tbinom{n+2}{4} - \tbinom{2}{4}\right],$$

$$\Sigma^1 n^3 = 8\left[\tbinom{n-2}{1} - \tbinom{-2}{1}\right] + 7\left[\tbinom{n-1}{2} - \tbinom{-1}{2}\right] + 6\left[\tbinom{n}{3} - \tbinom{0}{3}\right] + 6\left[\tbinom{n+1}{4} - \tbinom{1}{4}\right],$$

$$\Sigma^1 n^3 = 27\left[\tbinom{n-3}{1} - \tbinom{-3}{1}\right] + 19\left[\tbinom{n-2}{2} - \tbinom{-2}{2}\right] + 12\left[\tbinom{n-1}{3} - \tbinom{-1}{3}\right] + 6\left[\tbinom{n}{4} - \tbinom{0}{4}\right].$$

In the preceding example, some binomial coefficients have negative upper indices, which may look unfamiliar. These coefficients are well defined because negative upper-index binomial coefficients satisfy $\binom{-n}{k} = (-1)^k \binom{k+n-1}{k}$. Moreover, the binomial coefficient $\binom{n}{k}$ is a polynomial in n and therefore accepts fractional values in its upper index. Consequently, the expression in Proposition (3.1) is also a polynomial in n. The following example illustrates the polynomial form of the formula.

Example 3.3. *For integers t with $0 \le t \le 4$, we have*

$$
\begin{aligned}
t=0,\quad \Sigma^1 n^3 &= n\cdot 0 + \tfrac{1}{2}(n+n^2)\cdot 1 \\
&\quad + \tfrac{1}{6}(2n+3n^2+n^3)\cdot(-6) \\
&\quad + \tfrac{1}{24}(6n+11n^2+6n^3+n^4)\cdot 6, \\
t=1,\quad \Sigma^1 n^3 &= n\cdot 1 + \tfrac{1}{2}(-n+n^2)\cdot 1 \\
&\quad + \tfrac{1}{6}(-n+n^3)\cdot 0 \\
&\quad + \tfrac{1}{24}(-2n-n^2+2n^3+n^4)\cdot 6, \\
t=2,\quad \Sigma^1 n^3 &= n\cdot 8 + \tfrac{1}{2}(-3n+n^2)\cdot 7 \\
&\quad + \tfrac{1}{6}(2n-3n^2+n^3)\cdot 6 \\
&\quad + \tfrac{1}{24}(2n-n^2-2n^3+n^4)\cdot 6, \\
t=3,\quad \Sigma^1 n^3 &= n\cdot 27 + \tfrac{1}{2}(-5n+n^2)\cdot 19 \\
&\quad + \tfrac{1}{6}(11n-6n^2+n^3)\cdot 12 \\
&\quad + \tfrac{1}{24}(-6n+11n^2-6n^3+n^4)\cdot 6, \\
t=4,\quad \Sigma^1 n^3 &= n\cdot 64 + \tfrac{1}{2}(-7n+n^2)\cdot 37 \\
&\quad + \tfrac{1}{6}(26n-9n^2+n^3)\cdot 18 \\
&\quad + \tfrac{1}{24}(-50n+35n^2-10n^3+n^4)\cdot 6.
\end{aligned}
$$

The coefficients for the indicated values of t form the following OEIS sequences:

- For $t=0$, the coefficients $0, 1, -6, 6$ form sequence A278075 in the OEIS [8].
- For $t=1$, the coefficients $1, 1, 0, 6$ form sequence A389570 in the OEIS [8].
- For $t=2$, the coefficients $8, 7, 6, 6$ form sequence A391068 in the OEIS [8].
- For $t=3$, the coefficients $27, 19, 12, 6$ form sequence A391210 in the OEIS [8].
- For $t=4$, the coefficients $64, 37, 18, 6$ form sequence A395604 in the OEIS [8].

We derive a formula for double sums of powers similarly. Setting $r = 1$ in Lemma (1.6) yields the following result.

Proposition 3.4 (Double Sums of Powers)**.** *For non-negative integers n and m, and an arbitrary integer t, the following identity holds*

$$\Sigma^2 n^m = \sum_{k=0}^{m} \left[\binom{n-t+k+1}{k+2} - \binom{k-t+1}{k+2} \Sigma^0 n^0 - \binom{k-t+0}{k+1} \Sigma^1 n^0 \right] \nabla^k t^m.$$

The formula includes the correction terms $\Sigma^0 n^0 = 1$ and $\Sigma^1 n^0 = n$. The next example gives a formula for double sums of powers.

Example 3.5. *For integers t with $0 \le t \le 2$, we have*

$$\begin{aligned}
t = 0, \quad \Sigma^2 n^3 &= \tfrac{1}{2}(n + n^2) \cdot 0 + \tfrac{1}{6}(2n + 3n^2 + n^3) \cdot 1 + \tfrac{1}{24}(6n + 11n^2 + 6n^3 + n^4) \cdot (-6) \\
&\quad + \tfrac{1}{120}(24n + 50n^2 + 35n^3 + 10n^4 + n^5) \cdot 6, \\
t = 1, \quad \Sigma^2 n^3 &= \tfrac{1}{2}(n + n^2) \cdot 1 + \tfrac{1}{6}(-n + n^3) \cdot 1 + \tfrac{1}{24}(-2n - n^2 + 2n^3 + n^4) \cdot 0 \\
&\quad + \tfrac{1}{120}(-6n - 5n^2 + 5n^3 + 5n^4 + n^5) \cdot 6, \\
t = 2, \quad \Sigma^2 n^3 &= \tfrac{1}{2}(n + n^2) \cdot 8 + \tfrac{1}{6}(-4n - 3n^2 + n^3) \cdot 7 + \tfrac{1}{24}(2n - n^2 - 2n^3 + n^4) \cdot 6 \\
&\quad + \tfrac{1}{120}(4n - 5n^3 + n^5) \cdot 6.
\end{aligned}$$

Similarly, repeated application of Lemma (1.6) yields a closed formula for multifold sums of powers.

Theorem 3.6 (Multifold Sums of Powers)**.** *For non-negative integers r, n, and m, and an arbitrary integer t, the following identity holds*

$$\Sigma^r n^m = \sum_{k=0}^{m} \left[\binom{n-t+k+r-1}{k+r} - \sum_{s=0}^{r-1} \binom{k-t+r-1-s}{k+r-s} \Sigma^s n^0 \right] \nabla^k t^m.$$

The following example illustrates the multifold formula.

Example 3.7. *For non-negative integers r, n, and m, we have*

$$\Sigma^r\, n^m = \sum_{k=0}^{m} \left[\tbinom{n+k+r-1}{k+r} - \sum_{s=0}^{r-1} \tbinom{k+r-1-s}{k+r-s} \Sigma^s\, n^0 \right] \nabla^k 0^m,$$
$$\Sigma^r\, n^m = \sum_{k=0}^{m} \left[\tbinom{n+k+r-2}{k+r} - \sum_{s=0}^{r-1} \tbinom{k+r-2-s}{k+r-s} \Sigma^s\, n^0 \right] \nabla^k 1^m,$$
$$\Sigma^r\, n^m = \sum_{k=0}^{m} \left[\tbinom{n+k+r-3}{k+r} - \sum_{s=0}^{r-1} \tbinom{k+r-3-s}{k+r-s} \Sigma^s\, n^0 \right] \nabla^k 2^m,$$
$$\Sigma^r\, n^m = \sum_{k=0}^{m} \left[\tbinom{n+k+r-4}{k+r} - \sum_{s=0}^{r-1} \tbinom{k+r-4-s}{k+r-s} \Sigma^s\, n^0 \right] \nabla^k 3^m.$$

Using Lemma (1.7), we obtain the following binomial form of Theorem (3.6).

Proposition 3.8 (Multifold Binomial Sums of Powers)**.** *For non-negative integers r, n, and m, and an arbitrary integer t, the following identity holds*

$$\Sigma^r\, n^m = \sum_{k=0}^{m} \left[\binom{n-t+k+r-1}{k+r} - \sum_{s=0}^{r-1} \binom{k-t+r-1-s}{k+r-s} \binom{s+n-1}{s} \right] \nabla^k t^m.$$

Replacing t with $-t$ yields another formula for multifold sums of powers.

Proposition 3.9 (Multifold Binomial Sums of Powers Negated)**.** *For non-negative integers r, n, and m, and an arbitrary integer t, the following identity holds*

$$\Sigma^r\, n^m = \sum_{k=0}^{m} \left[\binom{n+t+k+r-1}{k+r} - \sum_{s=0}^{r-1} \binom{k+t+r-1-s}{k+r-s} \binom{s+n-1}{s} \right] \nabla^k (-t)^m.$$

The next example illustrates the formula after replacing t with $-t$.

Example 3.10. *For integers t with $1 \le t \le 4$, we have*

$$\begin{aligned}
t=1,\quad \Sigma^1 n^3 &= n\cdot(-1) + \tfrac{1}{2}(3n+n^2)\cdot 7 + \tfrac{1}{6}(11n+6n^2+n^3)\cdot(-12)\\
&\quad + \tfrac{1}{24}(50n+35n^2+10n^3+n^4)\cdot 6,\\
t=2,\quad \Sigma^1 n^3 &= n\cdot(-8) + \tfrac{1}{2}(5n+n^2)\cdot 19 + \tfrac{1}{6}(26n+9n^2+n^3)\cdot(-18)\\
&\quad + \tfrac{1}{24}(154n+71n^2+14n^3+n^4)\cdot 6,\\
t=3,\quad \Sigma^1 n^3 &= n\cdot(-27) + \tfrac{1}{2}(7n+n^2)\cdot 37 + \tfrac{1}{6}(47n+12n^2+n^3)\cdot(-24)\\
&\quad + \tfrac{1}{24}(342n+119n^2+18n^3+n^4)\cdot 6,\\
t=4,\quad \Sigma^1 n^3 &= n\cdot(-64) + \tfrac{1}{2}(9n+n^2)\cdot 61 + \tfrac{1}{6}(74n+15n^2+n^3)\cdot(-30)\\
&\quad + \tfrac{1}{24}(638n+179n^2+22n^3+n^4)\cdot 6.
\end{aligned}$$

4. Stirling Form

The following lemma relates Stirling numbers of the second kind $\left\{ {n \atop k} \right\}$ to backward differences of powers $\nabla^k t^m$.

Lemma 4.1 (Stirling Backward Differences)**.** *For non-negative integers m and k, and an arbitrary integer t, the following identity holds*

$$\nabla^k t^m = \sum_{j=k}^{m} \binom{t-k}{j-k} \left\{ {m \atop j} \right\} j!.$$

Proof. The identity $t^m = \sum_{k=0}^{m} \binom{t}{k} \left\{ {m \atop k} \right\} k!$, together with the binomial recurrence $\binom{n+1}{k} = \binom{n}{k} + \binom{n}{k-1}$, gives the result. □

The lemma gives the following representation.

Proposition 4.2 (Multifold Stirling Sums of Powers)**.** *For non-negative integers r, n, and m, and an arbitrary integer t, the following identity holds*

$$\Sigma^r n^m = \sum_{k=0}^{m} \sum_{j=k}^{m} \left[\binom{n-t+k+r-1}{k+r} - \sum_{s=0}^{r-1} \tbinom{k-t+r-1-s}{k+r-s} \tbinom{s+n-1}{s} \right] \binom{t-k}{j-k} \left\{ {m \atop j} \right\} j!.$$

5. Eulerian Form

The following lemma relates Eulerian numbers $\left\langle {n \atop k} \right\rangle$ to backward differences of powers $\nabla^k t^m$.

Lemma 5.1 (Eulerian Backward Differences)**.** *For non-negative integers m and k, and an arbitrary integer t, the following identity holds*

$$\nabla^k t^m = \sum_{j=0}^{m} \binom{t+j-k}{m-k} \left\langle {m \atop j} \right\rangle.$$

Proof. Worpitzky's identity $t^m = \sum_{k=0}^{m} \binom{t+k}{m} \left\langle {m \atop k} \right\rangle$, together with the binomial recurrence $\binom{n+1}{k} = \binom{n}{k} + \binom{n}{k-1}$, gives the result. See [9]. □

The lemma gives the following representation.

Proposition 5.2 (Multifold Eulerian Sums of Powers)**.** *For non-negative integers r, n, and m, and an arbitrary integer t, the following identity holds*

$$\Sigma^r n^m = \sum_{k=0}^{m} \sum_{j=0}^{m} \left[\binom{n-t+k+r-1}{k+r} - \sum_{s=0}^{r-1} \binom{k-t+r-1-s}{k+r-s} \binom{s+n-1}{s} \right] \binom{t+j-k}{m-k} \left\langle {m \atop j} \right\rangle.$$

6. Conclusions

In this manuscript, we derive closed formulas for multifold sums of powers of integers by combining the backward Newton interpolation formula with hockey-stick identities for binomial coefficients. We further obtain representations of multifold sums of powers in terms of Stirling numbers of the second kind and Eulerian numbers. Finally, we provide Wolfram Mathematica programs for the efficient verification of the derived identities.

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Mathematica Programs

Use the GitHub repository to validate the results using Mathematica programs. Table 1 lists the programs and the results they validate.

Mathematica Function	Validates or Prints
`ValidateBackwardNewtonsFormulaForPowers.txt`	Validates Lemma (1.4)
`ValidateOrdinarySumsOfPowers.txt`	Validates Proposition (3.1)
`ValidateDoubleSumsOfPowers.txt`	Validates Proposition (3.4)
`ValidateMultifoldSumsOfPowers.txt`	Validates Theorem (3.6)
`ValidateMultifoldBinomialSumsOfPowers.txt`	Validates Proposition (3.8)
`ValidateMultifoldBinomialSumsOfPowersNegated.txt`	Validates Proposition (3.9)
`ValidateMultifoldEulerianSumsOfPowers.txt`	Validates Proposition (5.2)
`ValidateMultifoldStirlingSumsOfPowers.txt`	Validates Proposition (4.2)

Table 1. Mathematica programs and the results they validate.

Metadata

- **Initial release date:** January 1, 2026.
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- **Version:** `2.1.0+main.ebec1f9`
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- **DOI:** https://doi.org/10.5281/zenodo.18118011
- **Web Version:** kolosovpetro.github.io/sums-of-powers-backward-differences
- **Sources:** github/kolosovpetro/SumsOfPowersBackwardDiffNewtonFormula
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URL: https://kolosovpetro.github.io